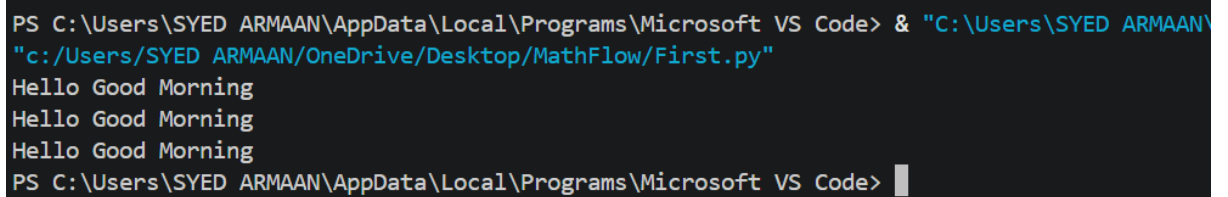


1.While Loop Example

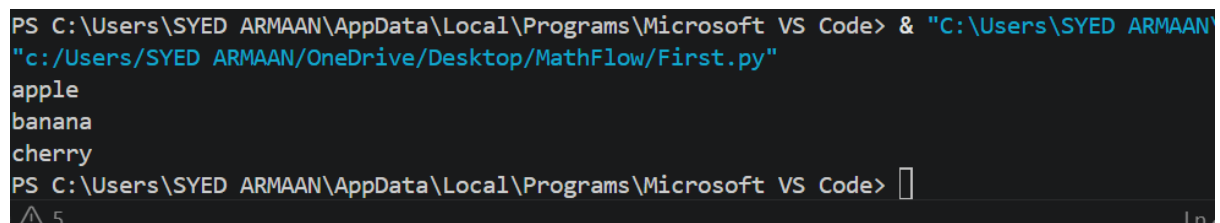
```
count = 0
while (count < 3):
    count = count + 1
    print("Hello Good Morning")
```

A terminal window with a dark background. The prompt is 'PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>'. The user enters '& "C:\Users\SYED ARMAAN\OneDrive\Desktop\MathFlow\First.py"'. The program outputs 'Hello Good Morning' three times, once on each line. The prompt returns.

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Hello Good Morning
Hello Good Morning
Hello Good Morning
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> 
```

2.For Loop Example

```
fruits = ["apple", "banana", "cherry"]
for fruit in fruits:
    print(fruit)
```

A terminal window with a dark background. The prompt is 'PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code>'. The user enters '& "C:\Users\SYED ARMAAN\OneDrive\Desktop\MathFlow\First.py"'. The program outputs 'apple', 'banana', and 'cherry' on three separate lines. The prompt returns.

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
apple
banana
cherry
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> 
```

3. Simple Calculator (Functions)

```
def add(a, b):  
    return a + b  
  
def multiply(a, b):  
    return a * b  
  
x = int(input("Enter first number: "))  
y = int(input("Enter second number: "))  
  
print("Sum:", add(x, y))  
print("Product:", multiply(x, y))
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\OneDrive\Desktop/MathFlow/First.py"  
Enter first number: 23  
Enter second number: 43  
Sum: 66  
Product: 989  
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> █
```

4. Temperature Checker (If-Else)

```
temp = int(input("Enter temperature: "))  
  
if temp > 30:  
    print("Hot")  
  
elif temp > 20:  
    print("Warm")  
  
else:  
    print("Cold")
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\OneDrive\Desktop/MathFlow/First.py"  
Enter temperature: 23  
Warm  
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\OneDrive\Desktop/MathFlow/First.py"  
Enter temperature: 33  
Hot  
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> █
```

5.Student Record (Dictionary)

```
students = {"Ali": 80, "Sara": 90}

name = input("Enter name: ")

if name in students:

    print("Marks:", students[name])

else:

    print("Student not found")
```

```
Enter name: armaan
Student not found
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Enter name: ali
Student not found
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Enter name: Ali
Marks: 80
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> |
```

6.Tuple Example

```
coordinates = (4, 5)

print(coordinates[0])

print(coordinates[1])
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
4
5
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> |
```

7. List Input Example

```
movies = ["Leo", "Jawan", "RRR"]

print("Available movies:", movies)

choice = input("Enter movie to book: ")

if choice in movies:

    print("Ticket booked for", choice)

    movies.remove(choice)

else:

    print("Movie not available")
```

```
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Available movies: ['Leo', 'Jawan', 'RRR']
Enter movie to book: alpha
Movie not available
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Available movies: ['Leo', 'Jawan', 'RRR']
Enter movie to book: Jawan
Ticket booked for Jawan
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> |
```

8. Classroom Attendance (Set)

```
students = set()

students.add("Ali")

students.add("John")

name = input("Enter name: ")

if name in students:

    print("Present")

else:

    print("Absent")
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
Enter name: armaan
Absent
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> |
```

9.String Slicing

```
s = "ABCDEF"

print(s[1:4])

print(s[:3])

print(s[3:])

print(s[::-1])
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
BCD
ABC
DEF
FEDCBA
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> |
```

10.Array

```
import array as arr

a = arr.array('i', [1, 2, 3])

print(a[0])

a.append(5)

print(a)
```

```
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\SYED ARMAAN\
"c:/Users/SYED ARMAAN/OneDrive/Desktop/MathFlow/First.py"
1
array('i', [1, 2, 3, 5])
PS C:\Users\SYED ARMAAN\AppData\Local\Programs\Microsoft VS Code> |
```

